



ACHIEVEMENTS AND SOLUTIONS IN MECHANICAL ENGINEERING III - THE 7th INTERNATIONAL CONFERENCE IN MECHANICAL ENGINEERING (ICOME)

**SELECTED PEER-REVIEWED FULL TEXT PAPERS FROM THE
7th INTERNATIONAL CONFERENCE IN
MECHANICAL ENGINEERING (ICOME 2024)**

**EDITED BY
ILIE DUMITRU
LAURENȚIU DANIEL RACILĂ
LUCIAN MATEI**



TRANS TECH PUBLICATIONS

Advances in Science and Technology

ISSN 1662- 8969

Editor in Chief

Dr. Farooq Sher, Nottingham Trent University, School of Science and Technology; Nottingham, United Kingdom, NG11 8NS;

Editorial Board: *see back inside cover.*

Aims and Scope:

Advances in Science and Technology (AST) includes peer-reviewed proceedings from Materials Science and Engineering related conferences and symposiums from all over the world. Published proceedings include engineering papers, review paper and papers about new materials research, development and application, its research, processing, and practical use in various areas of production and engineering. Papers submitted for publication in AST cannot be published or sent for publication elsewhere. Authors retain the right to publish an extended, significantly updated version in another periodical.

Abstracted/Indexed in:

Scopus
Google Scholar
NASA Astrophysics Data System (ADS)
ProQuest
Ulrichsweb
EBSCOhost Research Databases
Zetoc
Index Copernicus Journals Master List
WorldCat (OCLC)
CEABA
Poly
STEC

Internet:

The periodical is available in full text via www.scientific.net

ISSN print 1662-8969

ISSN web 1662-0356



Trans Tech Publications Ltd
Seestrasse 24c • CH-8806 Baech • Switzerland
Tel: +41 (44) 922 10 22 • e-mail: office@scientific.net
<https://www.scientific.net>

ISBN print: 978-3-0364-0941-2

ISBN eBook: 978-3-0364-1941-1

DOI book: 10.4028/b-7YVyXm

© 2026 Trans Tech Publications Ltd. All Rights Reserved

Achievements and Solutions in Mechanical Engineering III - The 7th International Conference in Mechanical Engineering (ICOME)

Selected peer-reviewed full text papers from the
7th International Conference in
Mechanical Engineering (ICOME 2024)

Edited by
Ilie Dumitru
Laurențiu Daniel Racilă
Lucian Matei

Achievements and Solutions in Mechanical Engineering III - The 7th International Conference in Mechanical Engineering (ICOME)

Selected peer-reviewed full text papers from the
7th International Conference in
Mechanical Engineering (ICOME 2024)

Selected peer-reviewed full text papers from the
7th International Conference in Mechanical Engineering (ICOME),
October 17-19, 2024, Craiova, Romania

Edited by

Ilie Dumitru, Laurențiu Daniel Racilă, Lucian Matei

■ **Scientific.Net**

Copyright © 2026 Trans Tech Publications Ltd, Switzerland

All rights reserved. No part of the contents of this publication may be reproduced, processed or transmitted in any form or by any means without the written permission of the publisher.

Trans Tech Publications Ltd
Seestrasse 24c
CH-8806 Baech
Switzerland
<https://www.scientific.net>

Volume 178 of
Advances in Science and Technology
ISSN print 1662-8969
ISSN web 1662-0356

Full text available online at <https://www.scientific.net>

Distributed worldwide by

Trans Tech Publications Ltd
Seestrasse 24c
CH-8806 Baech
Switzerland

Phone: +41 (44) 922 10 22
e-mail: sales@scientific.net

Table of Contents

Preface	v
Committee, Sponsors	vi
Chapter 1: Metal Processing Technologies	
Regarding the Use of Special Stamping Devices Obtained by Electrical Erosion in the Automobile Industry	
D. Popescu and I. Geonea	3
Temperature in Turning Process of Pure Titanium and Ti6Al4V Titanium Alloy	
D. Panduru, E.N. Patru, M. Bica and N. Craciunoiu	11
Results Regarding the Temperature in Milling Process of Aluminum Alloys and Stainless Steel	
E.N. Patru, D. Panduru, M. Bica and N. Craciunoiu	17
Chapter 2: Mechatronics, Machines and Equipment	
Study on Blade Geometric Design and Advanced Control for Horizontal Wind Turbine	
M.C. Călin, L. Matei, M. Muşat, M. Pandealea and M. Iliescu	25
Study on the Use of Robotic Arms in Logistics and Manufacturing: A Short Review	
M.L. Proistosescu Necsoi, E.L. Niţu and C.A. Gavriluţă	35
Manipulators Used as Endoscopic Instrumentation of the Urinary Tract	
O. Antonescu, V. Veliscu, I.D. Geonea and P. Antonescu	47
Critical Vibrations for Turbine Elements	
T. Ţunescu, I. Ştefan, B.A. Olei and I.D. Savu	55
Chapter 3: Conceptual Approaches in Automotive Engineering	
Researches in the Field of Electric and Hybrid Mobility at the Automotive Engineering Research Centre of the National University of Science and Technology Politehnica Bucharest - Piteşti Branch	
L. Călin, D.G. Marinescu and V. Nicolae	65
GAster Electricway 4WD Concept with Compressed Natural Gas, Continuously Variable Transmission and Electrified Rear Axle	
L. Călin, D.G. Marinescu and V. Nicolae	79
Constructive and Functional Optimisation of a Fuel Tank in the Supply System of a Spark Ignition Engine	
I. Dumitru, L. Racilă, G. Paraschiv, L. Matei and C. Drăghici	95
Enabling Technologies and System Architectures for Autonomous Vehicles	
T.G. Oprica, I. Dumitru, D. Burghilă, L. Simniceanu and L. Matei	105
A Short History of Electric Vehicles	
L. Racila, I. Dumitru, T. Oprica and L. Matei	115

Chapter 4: Living Conditions and Environmental Quality in Urban Areas	
Selection of Heat-Insulating Materials for the Thermal Rehabilitation of a Building	
M. Demian, G. Demian and L. Grecu.....	123
Selection of Materials for Noise Control in Buildings Using a Multicriteria Decision Method	
G. Demian, L. Grecu and M. Demian.....	133
Green Roofs for Office Buildings: Benefits to Mass Transfer and Energy Efficiency	
V. Iordan, M. Călbureanu, N.M. Marinache, A. Ionescu, D. Călbureanu and R. Malciu.....	143
Modelling of the Improvement of Living Conditions in the Apartment Blocks Built before 1989 and Rehabilitation through Thermal Insulation, Using Ubakus Heat and Mass Transfer through Walls Program	
A.M. Codiță, N.M. Marinache, A. Stăncuț and D.M. Călbureanu	155
The Importance of Using Appropriate Materials When Renovating Old Masonry Buildings - Case Study: Church from the End of the XIX-Th Century	
A.M. Codiță, N.M. Marinache, A. Stăncuț and A. Ionescu.....	161
Modelling and Processing of the Dynamic Modular Facades Manufactured by the U Bending Technology	
L.M. Ciurezu-Gherghe, C. Teisanu, B. Hurezeanu, M. Mamuleanu, G. Sima and O. Gingu.....	169
Developing and Using Some Indices for Studying the Evolution of Surface Water Quality	
L. Grecu, G. Demian and M. Demian.....	181
Keyword Index	191
Author Index	193



9 783036 409412

ISBN: 978-3-0364-0941-2
ADVANCES IN SCIENCE AND TECHNOLOGY VOL. 178
ELECTRONICALLY AVAILABLE AT WWW.SCIENTIFIC.NET